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Relationship of Systemic Cytokine Concentrations to Cognitive Function over Two Years in Women with Early Stage Breast Cancer

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Abstract

Cancer and its treatment are frequently associated with cancer-related cognitive impairment (CRCI). While CRCI has been linked to chemotherapy, there is increasing evidence that the condition may start prior to treatment and for some, remain unresolved after active treatment and into survivorship. Although the pathophysiology of the condition is complex, alterations in systemic cytokines, signaling molecules activated in response to infection or injury that trigger inflammation, are a possible mechanism linked to cognitive dysfunction in breast cancer and other

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Conflict of Interest

The authors declare that they have no conflict of interest.

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conditions. Given the conflicting results in the literature, the lack of focus on domain-specific cognitive testing, and the need for a longer time period given the multiple modalities of standard treatments for early-stage breast cancer, this longitudinal study was conducted to address these gaps.

Methods—We assessed 75 women with early-stage breast cancer at five points over two years, starting prior to the initial chemotherapy through 24 months after chemotherapy initiation. Measures included a validated computerized evaluation of domain-specific cognitive functioning and a 17-plex panel of plasma cytokines. Linear mixed-effects models were applied to test the relationships of clinical variables and cytokine concentrations to each cognitive domain. Results: Levels and patterns of cytokine concentrations varied over time: six of the 17 cytokines (IL-6, IL-12, IL-17, G-CSF, MIP-1 β , and MCP-1) had the most variability. Some cytokine levels (e.g., IL-6) increased during chemotherapy but then decreased subsequently, while others (e.g., IL-17) consistently declined from baseline over time. There were multiple relationships among cytokines and cognition, which varied over time. At baseline, elevated concentrations of G-CSF and reduced concentrations of IL-17 were associated with faster psychomotor speed. At the second time-point (prior to the mid-chemotherapy), multiple cytokines had significant associations with psychomotor speed, complex attention, executive function, verbal memory, cognitive flexibility, composite memory and visual memory. Six months after chemotherapy initiation and at the one-year point, there were multiple, significant relationships among cytokines and multiple cognitive. At two years, fewer significant relationships were noted; however, lower concentrations of IL-7, a hematopoietic cytokine, were associated with better psychomotor speed, complex attention, and memory (composite, verbal and visual). MCP-1 was inversely associated with psychomotor speed and complex attention and higher levels of MIP-1 β were related to better complex attention.

Conclusion—Levels and patterns of cytokines changed over time and demonstrated associations with domain-specific cognitive functioning that varied over time. The observed associations between cytokines and cognitive performance provides evidence that not only prototypical cytokines (i.e. IL-6, TNF- α , and IL-1 β) but also cytokines from multiple classes may contribute to the inflammatory environment that is associated with cognitive dysfunction. Future studies to better delineate the cytokine changes, both individually and in networks, are needed to precisely assess a mechanistic link between cytokines and cognitive function in women receiving treatments for breast cancer.

Background

Cancer-related cognitive impairment (CRCI) has been the subject of considerable research inquiry, with most research focused on post-chemotherapy CRCI in women with breast cancer. The pathophysiology of CRCI, although still unclear, is thought to involve the activation of cytokines, signaling molecules that mediate and regulate immunity, inflammation, and hematopoiesis. The link between inflammation and the development and progression of cancer is well-documented (Balkwill et al., 2005). Higher levels of circulating proinflammatory cytokines and their receptors have been found in individuals with cancer prior to treatment, during chemotherapy, and in survivorship. In addition to chemotherapy, cytokine interactions may be affected by other cancer treatments, including radiation therapy (Bentzen, 2006) and hormonal therapies, (Collins et al., 2009, Lee et al.,

2016) as well as multiple host factors including older age, (Hurria et al., 2006, Mandelblatt et al., 2016) menopausal status and symptoms, (Miura et al., 2016) and adiposity (Hartman et al., 2015). Although cytokines have been implicated in breast cancer development and progression for many years (Dethlefsen et al., 2013), the understanding of the multiple roles of cytokines in the central nervous system has been appreciated only in the more recent past (Maier, Goehler et al. 1998; Wilson, 2002). It is now well-accepted that cytokines can cross the blood brain barrier by active transport mechanisms in the choroid plexus and circumventricular organs, and have effects on neural processing (Dantzer et al., 2008), dopamine and serotonin metabolism, neural repair and neuronal/glia cell modulation (Aluise et al., 2010). Given these mechanistic linkages, there is mounting evidence that cytokines may influence neuroinflammation and contribute to cognitive and brain dysfunction in the context of breast cancer and its treatments (Janelsins et al., 2014).

The relationships among cytokine perturbations and cognitive outcomes is emerging across many fields (Wilson et al., 2002). Distinctive cytokine signatures have been identified in multiple immune-mediated medical conditions (e.g., heart failure (Pasic et al., 2003), Alzheimer's disease, (Swardfager et al., 2010), multiple sclerosis (Heesen et al., 2010), and Parkinson's (Dursun et al., 2015)). In our studies, we have shown that (Cohen et al., 2011) serum cytokine concentrations are associated with attention, executive function, and learning-memory performance among adults with HIV infection. Although a relationship between systemic cytokine concentrations and CRCI has been confirmed in animal models of cancer (Janelsins et al., 2011) and in some studies of women with breast cancer undergoing chemotherapy (Briones and Woods, 2014, Galimberti et al., 2006), past studies have tended to focus on "prototypical" cytokines or have been short-term studies that have not represented the potential cytokine changes in the context of multiple cancer treatments over time. A recent study using a multiplex panel of cytokines demonstrated multiple relationships with subjective cognitive impairment and cytokine networks, but the study was truncated to the first twelve weeks of the chemotherapy phase of treatment (Cheung et al., 2015a). In addition, the investigation of cytokines as a possible biological mechanism of cognitive dysfunction has been limited by the use of multiple measures of cognition (including subjective memory complaints) and cross-sectional or short-term longitudinal measures that focus on a single aspect of treatment, such as chemotherapy, radiation, hormonal therapy or survivorship. Further, most studies have measured prototypical cytokines that limits our understanding of pro- and anti-inflammatory interactive networks. To date, there have been very few studies that have investigated cognition over time, using objective neuropsychological batteries that have started prior to the initial chemotherapy and cytokine panels that permit the measure of multiple cytokines. Given the gaps in the literature, the lack of focus on domain-specific cognitive testing, and the need for a longer time period given the multi-modalities of standard treatments for early-stage breast cancer, we addressed these gaps by examining the association between plasma cytokine levels and cognitive performance across multiple domains in women with early-stage breast cancer over two years using validated, computerized cognitive testing.

Methods

As described in previous papers (Aboalela et al. 2015, Lyon et al., 2016), women between the ages of 21–65 with stage I to IIIA breast cancer were recruited from a National Cancer Center designated Cancer Center affiliated with an urban research University in the mid-Atlantic region and multiple collaborative sites across the state. A total of 77 women with early stage (I to IIIA) breast cancer, who ranged from 23 to 71 years of age, were recruited through 5 regional cancer centers in Central Virginia. Two women withdrew from the study prior to the initial data collection yielding an N=75. To identify potential study participants, each site had a study coordinator who screened patients for eligibility. The eligibility criteria were: (1) an age of 21 years or older; (2) a diagnosis of early stage breast cancer with a scheduled clinic appointment to receive chemotherapy; and (3) female gender (males were excluded since too few male participants were available for study). Exclusion criteria were a history of: (1) a previous cancer, or chemotherapy; (2) a diagnosis of dementia; (3) active psychosis; or (4) immune-related diagnoses (e. g. multiple sclerosis; systemic lupus erythematosus). After providing informed consent (VCU IRB #HM 13194), participants were enrolled and the first study visit was scheduled prior to the initiation of chemotherapy. The five time points for evaluation in this study were: prior to the start of chemotherapy (T1), at the midpoint of chemotherapy (T2), 6 months after the initial chemotherapy (T3), one year after the initial chemotherapy (T4) and two years after the initial chemotherapy (T5). The initial assessment (T1) was conducted after surgery but prior to commencing chemotherapy in women receiving adjuvant therapy. Women receiving neoadjuvant therapy had chemotherapy prior to surgical resection. After obtaining informed consent, participants were asked to complete questionnaires and performance-based cognitive testing via a computerized system. Participants were given incentives (a \$25 gift card to a local store that has both food and personal items for sale) at each data collection point over the two-year study.

Measures

Cytokines—Whole blood was collected at each study visit. Plasma was separated by centrifugation, and all specimens were aliquoted immediately, frozen, and stored in a -80°C freezer. A standard capture sandwich assay was used to determine the levels of different cytokines. Each captured antibody was coupled to a different bead set (Bio-Rad Laboratories, Hercules, California, USA). The system uses a liquid suspension array of 17 sets of 5.5- μm beads (Bio-Plex Human Cytokine 17-plex panel) internally dyed with different ratios of two spectrally distinct fluorochromes to assign a unique spectral address. Each set of beads was combined with a monoclonal antibody raised against interleukin (IL)-1 β , IL-2, IL-4, IL-5, IL-6, IL-7, IL-8, IL-10, IL-12 (p70), IL-13, IL-17, granulocyte-colony stimulating factor (G-CSF), granulocyte-macrophage colony stimulating factor (GM-CSF), interferon gamma (IFN γ), MCP-1 (monocyte chemotactic and activating factor), macrophage inflammatory protein (MIP-1 β) or tumor necrosis factor (TNF- α). Beads were incubated first (30 min to 2 h, at room temperature) with diluted standards (serial dilutions from 1.95 to 32 000 pg/ml) and then with biotinylated detector antibodies (30 min, at room temperature). They were washed twice in phosphate-buffered saline, and incubated for 30 min at room temperature with phycoerythrin-conjugated streptavidin. Each measurement

was taken in duplicate. Standard curves were generated by using the reference cytokine concentrations supplied by the manufacturer. Raw data (mean fluorescent intensity) were analyzed by Bio-Plex Manager Software (Bio-Rad Laboratories) to obtain concentration values. The lower limit of detection was <10 pg/ml (based on detectable signal >2 standard deviations (SD) above background). Samples were batch processed using plates from the same lots to reduce measurement variability.

Cognitive function—A performance-based computerized neurocognitive testing system, CNS Vital Signs™ (CNSVS, <https://www.cnsvs.com>) (Gualtieri and Johnson, 2006) was used to measure multiple cognitive domains. CNSVS was standardized with a normative sample of 1,069 subjects ranging in age from 7 to 90, drawn from the American population and has been used in multiple samples in research and clinical settings (Crawford and Jonassaint, 2015, Gualtieri and Johnson, 2006). Test results are presented in subject (raw) scores, age-matched standard scores, and percentile ranks. CNSVS standard scores have a mean of 100 and a standard deviation of 15; higher scores indicate better performance on memory and attention. The CNSVS scores individual tests and calculates a report of the clinical domains of neurocognitive functioning. The clinical domains include a neurocognitive index which is an average of five neurocognitive domains including: 1) memory, 2) psychomotor speed, 3) reaction time, 4) complex attention, and 5) cognitive flexibility. There are seven subtests that yield the five clinical domains and the combined neurocognitive index. The subscales of the CNSVS have good test-retest reliability: attention ($r = 0.65$), memory ($r = 0.66$), psychomotor speed ($r = 0.88$), cognitive flexibility ($r = 0.71$), and reaction time ($r = 0.75$) and have been used previously in women with breast cancer (Collins et al., 2015, Ercoli et al., 2015).

Covariates—Demographic, cancer and cancer treatment-related variables were collected by medical record review and participant interview. Demographic variables included: age, race, educational level, and body mass index. Body mass index (BMI) was calculated as weight divided by height squared. Cancer-related variables included breast cancer stage (TNM) and hormone receptor status. Treatment-related variables included surgery type, treatment regimen (adjuvant or neo-adjuvant), chemotherapeutic regimen, radiation status and hormonal agent status.

Statistical Analysis

All statistical analysis was performed in SAS version 9.4. Log-transformation was applied to the cytokine concentrations. To examine the cytokine log-concentrations changes across time, we fit linear mixed effects models with subject-specific random effects to account for the within subject correlation. Model-based F-tests were used to examine the overall temporal changes for each cytokines. To examine the relationship between cytokine log-concentrations and cognitive performance, we used backwards model selection procedure with the Akaike Information Criterion (AIC) (Akaike, 1974, Burnham and Anderson, 2002). The procedure is a linear regression model-selection algorithm that balances model fit with complexity to identify a parsimonious set of independent variables that predict a dependent variable. Demographic, cancer and cancer treatment-related variables were considered in the model selection as well.

Results

Clinical and demographic characteristics

Table 1 provides demographic and clinical characteristics of the cohort (Insert Table 1 here). All participants were adult females with a recent diagnosis of early-stage breast cancer between the ages of 23 and 71. All were scheduled for chemotherapy prior to study recruitment and were available for baseline evaluation prior the onset of their chemotherapy. The participants tended to be overweight based on their body mass index (mean BMI = 29.85 +7.47). Tumor stage and grade varied (Stage: I- 27%, IIA-41%, IIB-21%, IIIA-11%; Grade: 1-7%, 2-37%, 3-56%). The majority of women received *TAC* docetaxel (Taxotere) doxorubicin (Adriamycin), cyclophosphamide (Cytoxan) chemotherapy regimen. A majority (79%) had radiation therapy following chemotherapy and 44% were receiving hormonal therapies by T5.

Cytokine concentrations across time

Plasma concentrations of six of the 17 cytokines and chemokines (IL-6, IL-12, IL-17, G-CSF, MIP-1 β , MCP-1) changed significantly across visits. Figure 1 provides plots of the five cytokines in the sample across time. (Insert Figure 1 here). IL-6 and MIP-1 β concentrations increased from baseline through T3 (i.e., immediately post-radiation), and subsequently to levels that were below baseline. From very low baseline levels, MCP-1 concentrations also increased during chemotherapy. MCP-1 concentrations were reduced at 12- and 24-months relative to the chemotherapy period, but were still elevated relative to baseline. IL-12 and IL-17 concentrations were most elevated at baseline and showed a consistent decline across time. G-CSF was elevated at baseline and remained so through the initial chemotherapy period T2, but then declined precipitously over subsequent time-points.

Relationship of cytokine concentrations to cognitive function

Serum cytokine concentrations of specific cytokines were associated with cognitive performance on some, but not all domains of the CNSVS. The associations between cytokine concentrations and cognitive performance varied across time. Tables 2–6 detail the cytokines associated with the composite score for each cognitive domain, along with their beta coefficients.

Baseline cytokines and cognitive function (T1)—G-CSF concentrations were positively associated with cognitive flexibility ($t=2.33$, $p<.03$), executive functioning ($t=2.12$, $p<.04$), and psychomotor speed ($t=2.20$, $p<.04$), whereas IL-17 was negatively associated with psychomotor speed ($t = -2.69$, $p<.01$). GM-CSF was positively associated with reaction time ($t=2.52$, $p<.02$) at baseline. None of the other cytokines were associated with baseline performance in the domains of memory or complex attention (Insert Table 2 here).

Cytokines and cognitive function prior to mid chemotherapy (T2)—IL-7 concentrations were positively associated with composite memory ($t=2.41$, $p<.02$) and visual memory ($t=2.56$, $p<.02$). IL-5 and IL-17 were positively associated with psychomotor speed (IL-5: $t=2.18$, $p<.04$; IL-17: $t=4.94$; $p<.01$), whereas IL-1 β and IL-12 were negatively

associated with psychomotor speed (IL-1 β : $t=-3.19$, $p<.01$; IL-12: $t=-3.16$; $p<.01$). IL-4 and IL-1 β were differentially associated with complex attention (IL-4: $t=-2.16$, $p<.04$; IL-1 β : $t=2.21$, $p<.04$). IL-8 and IL-17 were differentially associated with cognitive flexibility (IL-8: $t=-3.21$; $p<.01$; IL-17: $t=2.80$, $p<.01$) and executive functioning (IL-8: $t=-3.20$; $p<.01$; IL-17: $t=2.71$, $p<.01$). IL-8 ($t=2.07$, $p<.05$) and IL-13 ($t=2.75$, $p<.01$) were positively associated with verbal memory whereas IL-12 ($t=-2.39$, $p=.02$) and IL-1 β ($t=-2.12$, $p<.04$) were inversely associated with verbal memory (Insert Table 3 here).

Cytokines and cognitive function post-chemotherapy (T3)—Multiple cytokines were significantly associated with cognitive performance post-chemotherapy (T3), though again this varied by cognitive domain. IL-17 was positively associated with psychomotor speed ($t=2.08$, $p<.05$) while IL-1 β was negatively associated with psychomotor speed ($t=-2.03$, $p<.05$). GM-CSF, IL-5, IL-7, and IL-12 were differentially associated with composite memory (GM-CSF: $t=-2.38$, $p=.02$; IL-7: $t=-2.40$; $p=.02$; IL-5: $t=2.11$; $p<.04$; IL-12: $t=2.14$, $p<.04$) at this time-point. Reaction time performance was differentially associated with G-CSF, IFN- γ , IL-7 and IL-17 concentrations (G-CSF: $t=-3.55$, $p<.01$; IFN- γ : $t=-3.48$, $p<.01$; IL-7: $t=2.50$; $p<.02$; IL-17: $t=2.67$, $p=.01$). IL-7 and IL-10 were differentially associated with both cognitive flexibility (IL-7: $t=2.99$; $p<.01$; IL-10: $t=-3.40$, $p<.01$) and executive function (IL-7: $t=3.11$; $p<.01$; IL-10: $t=-3.68$, $p<.01$). None of the cytokines were significantly associated with complex attention. G-CSF ($t=2.08$, $p<.05$) and IL-2 ($t=3.48$, $p<.01$) were positively associated with visual memory while IFN- γ ($t=-2.76$, $p<.01$) and IL-10 ($t=-2.91$, $p<.01$) were negatively associated with visual memory (Insert Table 4 here).

Cytokines and cognitive function one year post initial chemotherapy (T4)—Psychomotor speed was positively associated with MIP-1 β ($t=2.45$, $p<.02$) and negatively associated with MCP-1 ($t=-2.35$, $p<.03$). Reaction time was positively associated with IL-7 ($t=2.38$, $p=.02$) and negatively associated with IL-5 ($t=-2.05$, $p<.05$). IL-4 was positively associated with complex attention ($t=3.32$, $p<.01$) and cognitive flexibility ($t=2.59$, $p<.02$) while IL-6 was negatively associated with these cognitive domains ($t=-2.02$, $p<.05$; $t=-2.12$, $p<.04$ respectively). IFN- γ and IL-8 were negatively associated with executive functioning ($t=-2.17$, $p<.04$; $t=-2.94$, $p<.01$ respectively) while IL-4 was positively associated with executive functioning ($t=3.26$, $p<.01$) (Insert Table 5 here).

Cytokines and cognitive function at two years post initial chemotherapy (T5)—IL-7 was inversely associated with composite memory ($t=-2.75$; $p<.01$) and visual memory ($t=-2.99$, $p<.01$). GM-CSF and IL-17 were significantly associated with psychomotor speed (GM-CSF: $t=2.37$, $p<.03$; IL-17: $t=2.99$, $p<.01$) whereas MCP-1 and TNF- α were inversely associated with psychomotor speed (MCP-1: $t=-2.98$, $p<.01$, TNF- α : $t=-2.92$; $p<.01$). Complex attention was associated with MIP-1 β ($t=2.81$, $p<.01$) and inversely associated with MCP-1 ($t=-2.63$, $p<.02$). IL-5 was inversely associated with verbal memory ($t=-2.11$, $p<.04$) (Insert Table 6 here).

Discussion

This is one of the largest and longest cohort studies of serum cytokines and their relationship to performance across neurocognitive domains in women with early-stage breast cancer who

were assessed prior to chemotherapy and at targeted intervals over a two-year time period. Significant associations were found between serum cytokine and cognitive performance, though these associations varied as function of time, with concentrations of specific cytokines differing from pre-chemotherapy baseline through the chemotherapy treatment phase, and at subsequent long-term follow-up. Overall, these findings indicate that serum cytokine levels varied over time, corresponding to the pre-chemotherapy phase, chemotherapy, and to the recovery following multiple treatment modalities including chemotherapy and radiotherapy. Serum concentrations of six of the 17 cytokines and chemokines (IL-6, IL-12p70, IL-17, G-CSF, MIP-1 β , MCP-1) changed significantly over time (Figure 1). The trajectory of change differed among these six cytokines. IL-6, and the chemotactic cytokines MIP-1 β and MCP-1 increased from baseline through T3 (i.e., immediately post-chemotherapy), but then decreased subsequently over time to levels that were below baseline at the two-year follow-up. In contrast, IL-12 and IL-17 concentrations were very elevated at baseline, but then consistently declined across time, even during the chemotherapy treatment period. G-CSF was elevated at baseline and remained so through the initial chemotherapy period (T2) but then declined precipitously over subsequent time-points. The fact that these six cytokines showed three distinct trajectories of change suggests that they were likely responding differentially to the effects of cancer and cancer therapy (e.g. chemotherapy and radiation), and that these responses were affected by factors beyond the effects of chemotherapy alone.

Levels of IL-6 have been noted to be increased in women with breast cancer as compared to health controls and (Zhang and Adachi, 1999) (Benoy et al., 2002, Kozłowski et al., 2002). Likewise, serum levels of IL-6 (M = 111.38 pg/ml) as well as IL-12 (M = 1142.75 pg/ml) were significantly increased in breast cancer patients as compared to controls (Hussein et al., 2003). A recent study (Cheung, Ng, 2015a) also noted an increase in levels of IL-6 from prior to chemotherapy to six and 12 weeks after chemotherapy initiation. Likewise, serum IL-12 levels were significantly higher in patients than in controls, both without and after chemotherapy (Kovacs, 2001). Another study found IL-12 levels did not decrease after the initiation of anthracycline-based chemotherapy (Derin et al., 2007); however, the time-period for the study was over only two cycles of chemotherapy. Serum levels of IL-17 (median, 91.9 vs. 40.3 pg/ml, $p < 0.0001$), and G-CSF (median, 61.0 vs. 29.4 pg/ml, $p < 0.0001$) were significantly higher in breast cancer patients than in healthy volunteers and the levels of these markers were inversely correlated with disease outcomes (Kawaguchi et al., 2016, Schettler et al., 2014). In an in depth study of multiple chemokines, IL-8, MIP-1 alpha, MIP-1 β , were increased in 11 breast cancer patients compared to controls (Li et al., 2016).

Cognitive performance in the current cohort was associated with higher levels of some of the prototypical pro-inflammatory cytokines (e.g. IL-6) (Patel et al., 2015) that have been reported to be associated with cognitive dysfunction in other studies of women with breast cancer. Yet, other cytokines emerged that provide potentially novel associations with cognitive performance in this cohort such as IL-17, which has not been well studied in the context of CRCI. IL-17 was a significant correlate across multiple cognitive domains (executive functioning, cognitive flexibility, psychomotor speed). In addition to IL-17, IL-7 correlated with cognitive performance across four domains (reaction time, executive

functioning, cognitive flexibility, memory). IL-10 was associated with three cognitive domains (executive, cognitive flexibility, visual memory). Other cytokines were differentially associated with specific cognitive functions, but with less consistent and robust relationships.

Despite this evidence of an overall relationship between serum cytokine concentrations and reduced cognitive performance, the results indicate that these relationships are complex, as they vary both as a function of treatment stage, time, and cognitive domain. At baseline prior to chemotherapy, the relationship between the cytokines and cognitive function was largely limited to G-CSF, a cytokine that functions as a growth factor that stimulates the survival, proliferation, differentiation and function of neutrophil precursors and mature neutrophils and mobilizes hematopoietic stem cells from the bone marrow (Hollmén et al., 2016). A recent study, using a multi-stage mouse model of breast cancer, showed that granulocyte-colony stimulating factor (G-CSF) is the only hematopoietic growth factor to increase in serum during early tumor development (Casbon et al., 2015). G-CSF also has neurotropic effects in the brain, increasing neurogenesis and reducing apoptosis (Jeon et al., 2016, Schneider et al., 2005). In this sample, higher levels of G-CSF concentrations were positively associated with cognitive flexibility, executive functioning, psychomotor speed and reaction time at baseline, possibly reflecting its neurotropic properties. Although not well studied in the context of CRCI, G-CSF has been noted to have neurotrophic properties. Lower G-CSF plasma levels were found in 50 early Alzheimer's disease (AD) patients in comparison with 50 age-matched healthy controls (Laske et al., 2009). GM-CSF levels had a significant positive association with reaction time. GM-CSF, a putative growth factor for microglia/macrophages, has been associated with protective functions in a mouse model of AD in terms of Amyloid β plaque accumulation and cognitive function (Schettler, Klaidman, 2014). Similarly to a recent study of 99 women with early stage breast cancer using multiplex cytokine technology and computerized cognitive testing (Cheung et al., 2015b), higher concentration of IL-4 was associated with better cognitive performance in our study. In that study, higher IL-4 was associated with faster response in speed performance after the completion of the initial chemotherapy regimen ($P = 0.022$). IL-4 has a role in the regulation of brain immunity, with measureable downstream effects on spatial learning/memory and neurogenesis (Gadani et al., 2012). The proinflammatory cytokine, IL-17, had variable relationships with cognitive domains over time. While the observed associations between G-CSF, GM-CSF, and cognitive function are intriguing given their possible neuroprotective influence, recombinant human granulocyte colony-stimulating factor (r-metHuG-CSF; Filgrastim) G-CSF is a commonly prescribed adjunctive therapy in breast cancer patients for patients with neutropenia (Welte et al., 1996). Of note, G-CSF and GM-CSF which were positively associated with cognitive performance at baseline had the opposite relationship during the chemotherapy period (i.e., elevated concentrations corresponded with weaker cognitive performance). This relationship may have been influenced by the receipt of G-CSF-inducing medication Filgrastim (recombinant human granulocyte colony stimulating factor) by study participants with neutropenia. In these patients, higher levels could reflect the short-term effects of the medication instead of a stabilized level of endogenous G-CSF.

In contrast to relatively few significant relationships between cytokines and cognition at baseline, multiple cytokines were found to be associated with performance across cognitive

domains at T2 and T3, time-points that corresponded with the chemotherapy phase and immediately following completion of chemotherapy. Better executive function and cognitive flexibility, and faster reaction times were associated with higher levels of IL-7 (a growth factor involved in hematopoiesis and stem cell function) and IL-10 (an anti-inflammatory cytokine). These positive associations likely reflect beneficial effects of IL-7 and IL-10. However, the role of IL-10 in cognition is unclear. IL-10 was negatively associated with a composite score of executive function and processing speed in a sample of 1312 elders (Tegeler et al., 2016) and higher levels of IL-10 result in impaired neuronal differentiation in an animal model of the sub-ventricular zone (SVZ) of the lateral ventricle (LV), the main neurogenic area in the adult murine brain (Perez-Asensio et al., 2013).

The relationship between IL-17 concentrations and cognitive performance differed from the patterns observed for other cytokines and changed in direction over time. IL-17 was *negatively* associated with psychomotor speed at baseline, though the relationship was relatively modest. Yet, at all subsequent time-points, IL-17 was *positively* associated with cognitive performance. This relationship was most pronounced during the chemotherapy period (T2), as elevated IL-17 concentrations corresponded with better psychomotor speed, executive function, and cognitive flexibility. IL-17 is an important proinflammatory cytokine that may act as a bridge between early and late immune responses; this temporal factor may contribute to the varied relationships over time. Although little studied in CRCI, IL-17 levels have been associated with protective functions in a mouse model of Alzheimer's disease (AD) in terms of Amyloid β ($A\beta$) plaque accumulation and cognitive function (Pappu et al., 2011, Sun et al., 2015).

The relationships between the cytokines and cognitive performance at the one-year (T4) and two-year (T5) are of particular interest to the underlying mission of the current study. Since chemotherapy had been completed previously at T4, and for over 18 months at T5, measures from these two time-points reflect a long-term inflammatory status, free from active chemotherapy or radiotherapy treatments. Interestingly, at one year from chemotherapy initiation, a clearer pattern of relationships was noted for pro- and anti-inflammatory cytokines. There were commonalities but also some differences with respect to the cytokines that were associated with cognitive performance at these time points. For example, IL-7 continued to be positively associated with reaction time performance at T4, but was negatively associated with memory performance at T5. IL-17 was not associated with cognitive performance at T4, but was positively associated with psychomotor speed at T5. Other cytokines that were not implicated at earlier time points also emerged as correlates of cognitive performance at T4 and T5. At T4, MCP-1 and MIP-1 β were inversely associated with psychomotor speed. IL-4 and IL-6 were positively associated with complex attention and cognitive flexibility; while higher levels of IL-4, IL-8, and IFN- γ were associated with better executive functioning. None of the cytokines were associated with memory performance at T4.

At T5 (two years after initiation of chemotherapy), there were fewer relationships noted among pro- and anti-inflammatory cytokines and cognitive domains than at one-year post chemotherapy. At two years, the most notable associations between cognitive domains were with IL-7 and growth factors: levels of IL-7 were inversely related to composite and visual

memory; levels of MCP-I were inversely associated with psychomotor speed and complex attention; and, higher levels of MIP-1 β , which is proinflammatory but also related to homeostasis, were positively related to complex attention. MCP-1, one of the key chemokines that regulates migration and infiltration of monocytes/macrophages, including natural killer cells, remained elevated at 2-years post-breast cancer treatment. Although this finding is preliminary and warrants further study, elevated levels of MCP-1 have been noted in aging individuals with mild cognitive impairment compared to healthy controls (Galimberti, Fenoglio, 2006). TNF- α was also negatively associated with psychomotor speed at T5, whereas GM-CSF was positively associated with processing speed as was MIP-I β . IL-6, IL-8, and IFN- γ were inversely associated with multiple cognitive domains including reaction time, complex attention, cognitive flexibility, and executive function while cytokines IL-4, IL-5 and IL were positively related to reaction time complex attention, cognitive flexibility executive function.

It needs to be emphasized that the observed relationships between specific cytokine concentrations and performance in cognitive domains at one- and two- year follow-up occurred in the context of two broader longitudinal trends in the data. Most importantly, cognitive performance improved over time overall for women in the cohort. Therefore, relationships between cytokines and cognition at these points in recovery do not point to a sustained, deleterious effect of cancer treatments and decreased cognitive function in the cohort.

Conclusion

In this longitudinal study of women receiving chemotherapy for early-stage breast cancer, we observed heightened levels of proinflammatory cytokines IL-12, IL-17 and MIP-1 β , and G-CSF before chemotherapy. These cytokine levels fluctuated over the breast cancer treatment period and dropped precipitously at two-year post-breast cancer treatment. The observed associations between cytokines and cognitive performance provide evidence that not only prototypical cytokines (i.e. IL-6, TNF- α , and IL1- β) but also cytokines from multiple classes may contribute to the inflammatory environment that is associated with cognitive dysfunction. As many of these cytokines have been rarely assessed in prior studies, further research is warranted to determine their significance to contributing to cognitive performance in women undergoing treatment for early- stage breast cancer. Given the changes in cytokine levels over time and the relationships among cytokines and multiple domains of cognitive functioning, further study of these relationships is needed to confirm these relationships within the context of breast cancer treatment and survivorship.

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Highlights

Levels of proinflammatory cytokines IL-12, IL-17 and MIP-1 β , as well as G-CSF were elevated prior to chemotherapy.

Levels of six cytokines (IL-6, IL-12, IL-17, G-CSF, MIP-1 β , and MCP-1) varied markedly over time.

Cognitive outcomes in multiple domains were associated with cytokine levels with notable changes over time.

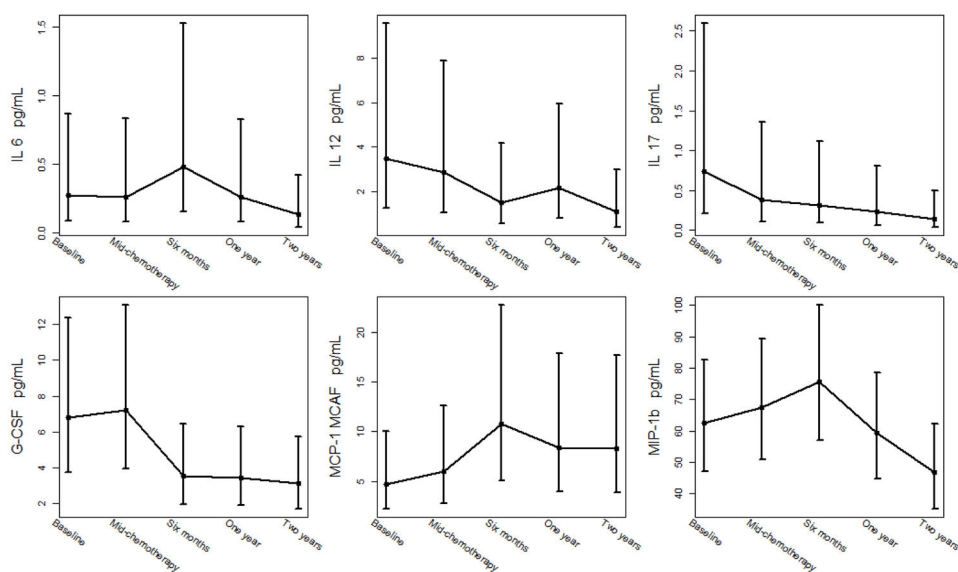


Figure 1.
Cytokine changes over time.

Table 1

Sample demographics and treatment characteristics

Variable	N = 75
Age (Mean (SD) [Range])	51.52 (10.34) [23.00, 71.00]
Race	
Caucasian	71% (53/75)
African-American	29% (22/75)
Other	0% (0/75)
Ethnicity	
Hispanic	4% (3/75)
Non-hispanic	96% (72/35)
Educational Level (%)	
Didn't finish High School	9% (7/75)
High School	12% (9/75)
Any Education Beyond High School	79% (59/75)
Employment (%)	
Unemployed	15% (11/75)
Disabled	8% (6/75)
Student	1% (1/75)
Part-Time	7% (5/75)
Full-time	55% (41/75)
Retired	15% (11/75)
Marital status	
Married/Partner	63% (47/75)
Divorced/Separated	24% (18/75)
Single Never Married	13% (10/75)
Household Income (%)	
Less than \$30,000	25% (19/75)
\$30,000–\$59,999	20% (15/75)
\$60,000–\$89,999	25% (19/75)
%90,000+	29% (22/75)
BMI (Mean (SD) [Range]))	29.85 (7.47) [19.11, 54.34]
Menopausal status (%)	
Pre- & Peri-menopause	43% (32/75)
Post-Menopause	57% (43/75)
Luminal A	
Yes	51% (38/75)

Variable	N = 75
No	49% (37/75)
Luminal B	
Yes	11% (8/75)
No	89% (67/75)
Triple negative	
Yes	29% (22/75)
No	71% (53/75)
HER2+, ER-& PR-	
Yes	9% (7/75)
No	91% (68/75)
Grade	
1	7% (5/75)
2	37% (28/75)
3	56% (42/75)
Stage	
I	27% (20/75)
IIA	41% (31/75)
IIB	21% (16/75)
IIIA	11% (8/75)
Surgery	
Biopsy	8% (6/74)
Lumpectomy	28% (21/74)
Segmental	20% (15/74)
Simple Mastectomy	43% (32/74)
Neoadjuvant	
Yes	11% (8/75)
No	89% (67/75)
Chemotherapy regimen	

Variable	N = 75
TAC	52% (39/75)
TC	28% (21/75)
TCH	14% (11/75)
CMF	3 % (2/75)
AC	3 % (2/75)
Radiation	
Yes	79% (59/75)
No	21% (16/75)
Hormonal Medication	
T4 Yes	26% (20/75)
T5 Yes	44% (33/75)

TAC docetaxel (Taxotere) doxorubicin (Adriamycin), cyclophosphamide (Cytosan); *TC* docetaxel (Taxotere), cyclophosphamide (Cytosan); *TCH* docetaxel (Taxotere), carboplatin (Paraplatin), trastuzumab (Herceptin); *CMF* cyclophosphamide, methotrexate, fluorouracil; *AC* doxorubicin (Adriamycin), cyclophosphamide (Cytosan)

Table 2

Time point 1 (T1)

Psychomotor Speed				Reaction Time			
Predictor	β	t Value	p-value	Predictor	β	t Value	p-value
BMI	1.0	-3.01	0.004	Education 1	32.0	-3.86	<0.001
Stage IIA	22.6	-3.60	0.001	Education 2	29.4	-4.41	<0.001
Stage IIB	24.2	-3.23	0.002	GM-CSF	1.0	-2.52	0.014
Stage IIIA	17.9	-2.00	0.049				
Neoadjuvant N	21.4	-2.43	0.018				
G-CSF	4.6	-2.20	0.031				
IL-17	2.8	-2.69	0.009				
Complex Attention				Cognitive Flexibility			
Education 1	24.9	-2.65	0.010	Race	14.5	-2.95	0.004
Education 2	25.8	-3.38	0.001	G-CSF	2.6	-2.33	0.023
Executive Functioning				Composite Memory			
Race	14.3	-3.02	0.004	Surgery Biopsy	19.0	-3.28	0.002
G-CSF	2.3	-2.12	0.037	Surgery Lumpectomy	8.5	-2.32	0.023
				Surgery Segmental	13.7	-3.29	0.002
Verbal Memory				Visual Memory			
Race	11.8	-2.92	0.005	Surgery Biopsy	17.1	-2.85	0.006
				Surgery Lumpectomy	7.5	-1.99	0.051
				Surgery Segmental	13.2	-3.06	0.003

Table 3

Time point 2 (T2)

Psychomotor Speed				Reaction Time			
Predictor	β	t Value	p-value	Predictor	β	t Value	p-value
Education 1	49.0	-4.88	<0.001	Education 1	21.8	-2.74	0.008
Education 2	52.3	-6.78	<0.001	Education 2	19.9	-3.24	0.002
Surgery Biopsy	24.6	-2.01	0.049				
Surgery Lumpectomy	14.3	-2.63	0.011				
Surgery Segmental	12.3	-1.88	0.066				
Neoadjuvant N	26.3	-2.37	0.021				
IL-5	2.0	-2.18	0.033				
IL-12	2.8	-3.16	0.002				
IL-17	4.5	-4.94	<0.001				
IL- β	3.6	-3.19	0.002				
Complex Attention				Cognitive Flexibility			
IL-4	2.6	-2.16	0.034	Race	11.2	-2.15	0.035
IL-1 β	2.5	-2.21	0.031	IL-8	4.0	-3.21	0.002
				IL-17	1.8	-2.80	0.007
Executive Functioning				Composite Memory			
Race	11.0	-2.12	0.038	Race	13.8	-3.87	<0.001
IL-8	4.0	-3.20	0.002	IL-7	1.6	-2.41	0.019
IL-17	1.7	-2.71	0.009				
Verbal Memory				Visual Memory			
BMI	0.5	-2.16	0.035	Race	11.6	-3.03	0.003
Surgery Biopsy	0.5	-0.08	0.935	IL-7	1.8	-2.56	0.013
Surgery Lumpectomy	2.6	-0.62	0.540				
Surgery Segmental	16.4	-3.60	0.001				
IL-8	2.3	-2.07	0.042				
IL-12	1.6	-2.39	0.020				
IL-13	2.1	-2.75	0.008				

Psychomotor Speed				Reaction Time			
Predictor	β	<i>t</i> Value	<i>p</i> -value	Predictor	β	<i>t</i> Value	<i>p</i> -value
IL-1 β	1.8	-2.12	0.038				

Table 4

Time point 3 (T3)

Psychomotor Speed				Reaction Time			
Predictor	β	t Value	p-value	Predictor	β	t Value	p-value
Education 1	28.4	-2.25	0.028	Surgery Biopsy	16.9	-1.79	0.078
Education 2	33.1	-3.33	0.001	Surgery Lumpectomy	1.5	-0.41	0.686
IL-17	2.3	-2.08	0.041	Surgery Segmental	15.4	-3.76	<0.001
IL-1 β	2.9	-2.03	0.047	Neoadjuvant N	21.1	-2.32	0.024
				G-CSF	3.6	-3.45	0.001
				IFN- γ	2.4	-3.48	0.001
				IL-7	2.5	-2.50	0.015
				IL-17	1.5	-2.67	0.010
Complex Attention				Cognitive Flexibility			
Education 1	23.0	-2.83	0.006	BMI	0.6	-2.27	0.027
Education 2	17.5	-2.57	0.012	Race	16.2	-3.81	0.000
Race	11.3	-2.49	0.015	Chemo AC	28.6	-1.64	0.106
				Chemo CMF	29.1	-2.33	0.023
				Chemo TAC	0.5	-0.10	0.921
				Chemo TC	15.0	-2.46	0.017
				IL-7	3.3	-2.99	0.004
				IL-10	2.2	-3.40	0.001
Executive Functioning				Composite Memory			
BMI	0.6	-2.35	0.022	Education 1	18.9	-2.53	0.014
Race	16.6	-3.96	<0.001	Education 2	23.8	-3.79	<0.001
Chemo AC	27.1	-1.58	0.119	Race	19.8	-4.71	<0.001
Chemo CMF	30.9	-2.51	0.015	GM-CSF	1.3	-2.38	0.020
Chemo TAC	1.0	-0.18	0.858	IL-5	1.8	-2.11	0.039
Chemo TC	15.5	-2.59	0.012	IL-7	2.5	-2.40	0.020
IL-7	3.4	-3.11	0.003	IL-12	1.2	-2.14	0.036
IL-10	2.3	-3.68	0.001				

Psychomotor Speed				Reaction Time			
Predictor	β	t Value	p-value	Predictor	β	t Value	p-value
Verbal Memory				Visual Memory			
Race	18.0	-4.64	<0.001	BMI	0.8	-3.42	0.001
				Race	18.8	-4.69	<0.001
				Chemo AC	18.6	-1.20	0.237
				Chemo CMF	43.2	-3.57	0.001
				Chemo TAC	8.4	-1.66	0.102
				Chemo TC	17.5	-3.09	0.003
				G-CSF	2.4	-2.08	0.042
				IFN- γ	3.2	-2.76	0.008
				IL-2	3.5	-3.48	0.001
				IL-10	2.7	-2.91	0.005

Table 5

Time point 4 (T4)

Psychomotor Speed				Reaction Time			
Predictor	β	t Value	p-value	Predictor	β	t Value	p-value
Education 1	39.42	-3.73	<0.001	Education 1	20.05	-2.96	0.004
Education 2	44.93	-5.24	<0.001	Education 2	18.28	-3.36	0.001
BMI	0.75	-2.22	0.030	IL-5	1.33	-2.05	0.044
Neoadjuvant N	24.22	-2.82	0.006	IL-7	2.59	-2.38	0.020
MCP-1	2.62	-2.35	0.022				
MIP-1 β	7.67	-2.45	0.017				
Complex Attention				Cognitive Flexibility			
Education 1	11.21	-1.34	0.185	Education 1	21.23	-2.47	0.016
Education 2	19.01	-2.82	0.007	Education 2	26.16	-3.76	<0.001
IL-4	2.57	-3.32	0.002	IL-4	2.06	-2.59	0.012
IL-6	1.53	-2.02	0.048	IL-6	1.66	-2.12	0.038
Executive Functioning				Composite Memory			
Education 1	25.33	-3.15	0.003	Education 1	14.30	-1.74	0.087
Education 2	25.45	-3.90	<0.001	Education 2	24.13	-3.67	0.001
Stage IIA	10.27	-2.11	0.039				
Stage IIB	13.98	-2.38	0.021				
Stage IIIA	25.16	-3.34	0.002				
Surgery Biopsy	7.38	-0.97	0.338				
Surgery Lumpectomy	16.18	-3.38	0.001				
Surgery Segmental	2.43	-0.45	0.652				
IFN- γ	2.33	-2.17	0.034				
IL-4	3.62	-3.26	0.002				
IL-8	4.61	-2.94	0.005				
Verbal Memory				Visual Memory			
BMI	0.73	-2.66	0.010	Neoadjuvant N	13.56	-2.11	0.038
Race	12.91	-2.83	0.006				

Table 6

Time point 5 (T5)

Psychomotor Speed				Reaction Time			
Predictor	β	t Value	p-value	Predictor	β	t Value	p-value
Education 1	26.1	-2.02	0.048	Education 1	24.3	-2.78	0.007
Education 2	38.5	-3.65	0.001	Education 2	22.6	-3.16	0.002
Neoadjuvant N	44.0	-4.29	<0.001				
GM-CSF	1.8	-2.37	0.021				
IL-17	2.4	-2.99	0.004				
MCP-1	4.3	-2.98	0.004				
TNF- α	2.8	-2.92	0.005				
Complex Attention				Cognitive Flexibility			
Education 1	26.4	-2.98	0.004	Education 1	26.1	-2.73	0.008
Education 2	28.1	-3.91	<0.001	Education 2	27.1	-3.45	0.001
Neoadjuvant N	14.6	-2.04	0.046				
MCP-1	2.8	-2.63	0.011				
MIP-1 β	10.7	-2.81	0.007				
Executive Functioning				Composite Memory			
No predictor was selected				BMI	0.7	-2.61	0.011
				Race	16.3	-3.49	0.001
				Neoadjuvant N	22.1	-2.79	0.007
				IL-7	3.0	-2.75	0.008
Verbal Memory				Visual Memory			
Race	19.5	-4.33	<0.001	Education 1	15.7	-1.83	0.072
IL-5	1.3	-2.11	0.039	Education 2	30.2	-4.31	<0.001
				Neoadjuvant N	21.8	-3.08	0.003
				IL-7	3.0	-2.99	0.004

Note for Tables 2–6:

Education: Education 2 is any education beyond High School; Education 1 is high school graduate; Education 0 is did not complete high school. Education 1 is the reference group.
 Neoadjuvant: Neoadjuvant N is participants who did not receive neoadjuvant chemotherapy; Neoadjuvant Y: participants who received neoadjuvant chemotherapy.

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Neoadjuvant N is the reference group.
Race: Caucasian is the reference group.
Surgery types: Simple mastectomy surgery is the reference group.
Stage of Cancer: Stage 1 is the reference group.
Chemotherapy Regimen: TCH is the reference group.

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